LUMENCORE

LUMAARC BRAND MARK

MARKET RESEARCH CAPABILITY STATEMENT

Project Argos Agentic AI Proof of Concept

Sources Sought Notice ONC-ARGOS-SSN-2026-OS351107

DRAFT - HUMAN REVIEW AND ACTION-TIME FACTS REQUIRED

Required cover field	Current response
Company name	LumenCore
Responding legal entity	Pending action-time fact
UEI / DUNS if applicable	Pending action-time fact
Company address	Pending action-time fact
Authorized point of contact	Pending action-time fact
Telephone / email	Pending action-time fact
Small-business designation(s)	Pending action-time fact
Response deadline	July 30, 2026 at 5:00 PM Eastern

Market research response only. No proprietary, classified, confidential, CUI, patient, or sensitive information is included.

1. Executive Fit and Recommended Role

Based on named first-party software artifacts, LumenCore can demonstrate source-artifact custody, versioned rule execution, evidence-package assembly, human approval gates, adverse-result retention, and reproducible reviewer handoff. Evidence packages are accompanied by SHA-256 integrity manifests. These are bounded component capabilities, not comparable federal-health past performance.

For Project Argos, those patterns align most directly with requirements traceability, public-source ingestion provenance, deterministic validator orchestration, evidence-case generation, corrective-action retest records, and production handoff documentation. LumenCore is responding at component level and does not claim presently qualified full-scope health IT prime readiness, FHIR R4 conformance authority, HHS ATO leadership, or an evaluated health-domain deployment.

Response posture: bounded small-business component capability. LumenCore is the sole respondent; no outside team is proposed; full-prime readiness is not claimed.

2. Understanding of the Requirement

- Argos is a governed monitoring and evidence system, not a general-purpose chatbot. Standards-based and legal pass/fail decisions should remain deterministic wherever possible, with AI limited to bounded support such as summarization, classification, reconciliation, and drafting.
- The evidence chain begins with authoritative public sources and must preserve raw artifacts, request context, timestamps, response metadata, normalized records, lineage decisions, validation outputs, and hashes.
- Suspected non-conformities require human review, rule-level traceability, deduplication, plain-language narratives, controlled corrective-action support, and before/after retesting.
- Real-world endpoint observation must be safe and unauthenticated where appropriate, must avoid patient data and protected health information absent separate written authorization, and must operate within approved rate, scope, security, and hosting boundaries.
- Production readiness includes Government-aligned deployment, operations and rollback runbooks, cost and staffing visibility, and an HHS authorization path; a proof-of-concept result alone is not an Authority to Operate.

3. Task-by-Task Capability and Gap Matrix

The matrix distinguishes current first-party component capability from work that has not been demonstrated. It does not assign any task to an unnamed partner or represent another organization's authority.

Task	Requirement	Proposed position	Bounded contribution
1	Initiation, requirements, and regulatory traceability	Partial capability	Build a versioned requirements-to-test-to-evidence matrix with evidence IDs, severity, human checkpoints, schedules, and change control. No complete 45 CFR or CHPL mapping or authorized regulatory interpretation is claimed.
2	Monthly administrative reporting	Partial capability	Produce source-backed status, blockers, milestones, event timelines, evidence IDs, decisions, and unresolved gates. Comparable HHS reporting performance is not claimed.
3	Public-source discovery and ingestion	Partial capability	Implement source inventory, raw-response custody, timestamps, headers, hashes, lineage, precedence, dictionaries, and normalized projections. CHPL, Lantern, NPPES, and FHIR connectors are not demonstrated.

Task	Requirement	Proposed position	Bounded contribution
4	Agentic architecture and governance	Partial capability	Separate deterministic validators from AI support; version policies, prompts, rules, retries, fallbacks, and audit logs; require human compliance decisions. No deployed health-domain architecture or secure HHS boundary is demonstrated.
5	Publication and FHIR R4 conformance testing	Not demonstrated	Package supplied validator outputs as traceable evidence cases and regression receipts. No FHIR R4 test authority, quarterly operation, publication-failure detection, or conformance acceptance is claimed.
6	Real-world endpoint observation	Not demonstrated	Contribute safe-observation logs, scope controls, drift records, hashes, and replay. Health-endpoint semantics, capability-sufficiency tests, published-versus-observed comparison, and approved access are not demonstrated.

Task	Requirement	Proposed position	Bounded contribution
7	Evidence case generation and triage	Strongest partial capability	Assemble raw and normalized artifacts, logs, timestamps, hashes, severity, rule traceability, narratives, deduplication, screenshots, clustering, and human/machine exports. A designated HHS-system export remains unproven.
8	Corrective-action workflow support	Partial capability	Provide controlled drafting, review, authorization, retest, before/after evidence, and recurrence fields. No ONC/ONC-ACB discretion or regulatory interpretation authority is claimed.
9	HHS Authority to Operate	Not demonstrated	Supply component inventories, reproducibility and change receipts, evidence IDs, and control-test artifacts. No HHS ATO lead, FIPS 199/SSP delivery, 3PAO coordination, milestone reporting, or authorization package is claimed.

Task	Requirement	Proposed position	Bounded contribution
10	Production-ready PPC release package	Partial capability	Package code, dependency locks, manifests, runbooks, rollback, limitations, tests, assumptions, costs, staffing, and reviewer receipts. No HHS-deployable package, hosting approval, or production acceptance is demonstrated.
11	Public-artifact assessment and FY 2027 strategy	Partial capability	Apply traceability to approved public-document checks and locked prototype scoring. EHI/API criteria, controlling regulatory citations, and a validated FY 2027 strategy are not demonstrated.

4. Proposed Evidence-Assurance Workstream

LumenCore proposes a fail-closed sequence in which missing authority, provenance, validation, or human disposition remains visible rather than becoming a promoted compliance claim.

1. Authorize sources and scope: Record approved source, purpose, collection boundary, cadence, rate limits, and prohibited data before collection.

2. Preserve raw observations: Store source payloads and request context with UTC timestamps, immutable identifiers, and SHA-256 manifests.

3. Normalize with lineage: Create common records while retaining source precedence, conflicts, and reversible links to raw artifacts.

4. Run deterministic checks: Apply versioned rules and validators; retain rule IDs, inputs, outputs, failures, and environment details.

5. Use AI only inside policy: Permit bounded summaries, clustering, reconciliation, and draft narratives; block autonomous compliance determinations.

6. Assemble an evidence case: Package artifacts, logs, hashes, timestamps, severity, traceability, narrative, and unresolved questions in human- and machine-readable forms.

7. Require human disposition: Route suspected issues to authorized reviewers and record decisions, rationale, corrective-action drafts, and retest outcomes.

5. Component Evidence and Similar-Scope Status

The following named first-party records support the engineering patterns offered here and are available for authorized review after repository security screening. They are not presented as healthcare, agency, field, or economic validation.

Evidence record	What the record supports	What it does not support
First-party reproducibility package	A named package and SHA-256 receipt record successful execution of its declared checks under an identified source state.	First-party bounded reproducibility only; not external validation, agency certification, field performance, or health IT past performance.
Custody and validation controls	Versioned manifests, SHA-256 receipts, schema checks, duplicate-action locks, and fail-closed gate records support an inspectable evidence workflow.	Control artifacts do not establish health-domain correctness, production authorization, contract performance, or agency acceptance.
Adverse-result retention	Named first-party records preserve failed promotion gates, negative findings, and unresolved authorities instead of converting them into favorable claims.	Transparent failure handling is an engineering pattern, not proof of FHIR conformance, regulatory interpretation, or field performance.

The notice requests experience of similar scope and complexity. The matrix below distinguishes LumenCore component evidence from material qualifications that are not demonstrated; it does not substitute adjacent technical work for federal health prior performance.

Capability area	Present support	Acquisition implication
Evidence custody, traceability, and deterministic validation	Component pattern supported by named first-party code, tests, receipts, and reviewer artifacts.	Available for authorized review after repository security screening.

Capability area	Present support	Acquisition implication
FHIR R4, CHPL/Lantern/NPPES, and ONC Certification Program delivery	No direct LumenCore prior-performance reference is claimed.	Future acquisition staffing or separately authorized teaming would be required; no outside team is proposed in this response.
HHS ATO, FIPS 199, SSP/control implementation, and 3PAO coordination	No direct LumenCore HHS authorization reference is claimed.	Future acquisition staffing or separately authorized teaming would be required; no ATO lead or assessor is proposed in this response.
Full-scope federal health program integration	No full-prime readiness or comparable federal health delivery is claimed.	LumenCore should not be treated as a presently qualified full-scope prime.

The internal evidence index states the present boundary directly: current strengths are artifact custody, deterministic replay, fail-closed claim governance, reviewer handoff packaging, and bounded pilot design. Independent scientific validation, field-validated savings, agency endorsement, certified safety, audited revenue, and customer adoption require separately identified external records.

6. Delivery and Security Approach

- Operate in a Government-approved environment with explicit authorization boundaries, least privilege, authenticated administration, dependency locking, secrets separation, immutable audit records, and rollback procedures.
- Treat HHS ATO as a managed authorization program: FIPS 199 categorization, boundary definition, SSP/control implementation, evidence collection, assessment coordination, POA&M handling, and Authorizing Official review.
- Do not collect, request, store, or use patient data or PHI unless separately authorized in writing and supported by approved architecture and controls.
- Keep AI components advisory and inspectable. Every model, prompt, policy, validator, retry, fallback, and rule version must be traceable to the evidence case it influenced.

Current boundary: the cited first-party artifacts are not an HHS-authorized production system and do not establish an ATO, FHIR certification, 3PAO assessment, or federal health deployment. Those remain acquisition and authorization gaps.

7. Standalone Response and Capability Gaps

Area	Required or relevant capability	Current status
Response posture	LumenCore is the sole respondent. No teaming arrangement, joint venture, or subcontracting relationship is proposed.	No external proposed team members exist to identify.
Health IT/FHIR and ONC program authority	CHPL/Lantern/NPPES semantics, FHIR R4 Endpoint/Organization/Bundle testing, ONC regulatory mapping, corrective-action content review.	Capability gap disclosed; no person or organization is represented.
Federal cybersecurity/ATO authority	HHS authorization boundary, FIPS 199, SSP, control implementation, assessment coordination, POA&M and authorization package.	Capability gap disclosed; no person or organization is represented.
Comparable full-program delivery	Contract performance, staffing, Government coordination, hosting, delivery acceptance, health-domain prior performance, and integrated schedule.	Not demonstrated and not claimed.

Teaming status as of submission: LumenCore has not formed or been authorized to represent a team for this response. No other organization or individual is proposed as a team member. The rows above identify capability gaps, not proposed or committed team members. LumenCore is open to future acquisition or teaming discussions, but this response does not represent another organization's commitment, qualifications, past performance, or authority.

8. Illustrative Mobilization Plan

The schedule below is an acquisition-planning outline, not a binding offer. It must be reconciled with Government scope, hosting, staffing, review cadence, security boundary, and the SOW's authorization milestones.

Period	Illustrative outcome
0-30 days	Confirm scope, source authorization, regulatory ownership, hosting assumptions, traceability schema, security boundary, and pilot cohort.
31-90 days	Stand up bounded ingestion, lineage, deterministic validator interfaces, evidence-case schema, human work queue, baseline test corpus, and audit logs.
91-180 days	Expand approved endpoint observation, drift tracking, issue clustering, corrective-action draft/retest workflow, security documentation, and operational runbooks.
181 days and beyond	Complete Government-directed hardening, assessment evidence, production handoff, PPC evaluation, cost/staffing analysis, and FY 2027 strategy.

9. Questions and Requested Next Step

1. What proof-of-concept duration, target cohort size, and Government hosting environment should respondents assume for acquisition planning?

2. Which organization owns final interpretation of 45 CFR and ONC Certification Program test outcomes, and what review service levels are expected?

3. Will HHS provide an approved source list, rate-limit policy, synthetic test corpus, and expected evidence-case export schema?

4. Which deliverables define the minimum viable first acquisition phase, and may HHS evaluate a bounded small-business contribution separately from full-scope prime readiness?

5. What existing HHS authorization boundary, reusable controls, continuous-monitoring services, or 3PAO arrangements may be available to the PPC?

For acquisition planning, HHS may treat this response as evidence of bounded small-business capacity in evidence management and validation orchestration. If HHS conducts follow-up market research, LumenCore is available to demonstrate the cited artifacts and answer technical questions.

This response is for market research only and is not an offer, proposal, certification, or representation that every draft SOW task is presently covered.